Roll No.: _

Amrita Vishwa Vidyapeetham, Coimbatore Campus

B.Tech Mid-Term Examinations – August 2025

Fifth Semester

Department of Computer Science and Engineering

23CSE304 Embedded Systems

Duration: Two hours Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Understand the fundamentals, functions, and architecture of embedded systems.
CO02	Develop applications making use of GPIOs and peripheral units using Embedded C.
CO03	Develop applications for the usage of Timers, Interrupts, and Debugging of embedded applications.
CO04	Design an embedded system using an ARM processor.

Answer all questions

PART A (Theory)[20 Marks]

1) Compare a microcontroller and a microprocessor in terms of their components using the use case of a smart home and a personal computer. [2] [CO01][BTL 2]

2) You have been hired as a junior embedded systems engineer at a company that develops smart wearable health devices (like fitness bands or smartwatches). Your first task is to understand how the internal embedded system of such a device is structured so you can later help optimize its design. Draw the basic architecture of an embedded system and explain the function of each major component (such as processor, memory, sensors, actuators, and communication interfaces) in the context of a wearable health monitoring device.

[4] [CO01][BTL 2]

3) **Answer the following**
   a. Differentiate between Thread mode and Handler mode in ARM Cortex-M processors. Which of these modes is used exclusively for privileged operations?
   b. Can the system switch from unprivileged mode to privileged mode using software? Explain briefly. [3] [CO01][BTL 2]

4) Explain the significance of the following condition flags found in CPSR: N (Negative), Z (Zero), C (Carry), V (Overflow) in the ARM Cortex-M4 processor. How do these flags affect the operation of the processor during program execution?

[3] [CO01][BTL 2]

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5)

a. Apply the concept of bit-banding in ARM Cortex-M4 to control a GPIO pin using STM32F401CCU6.
   i. Calculate the bit-band alias address for bit 5 of the register located at 0x4002001C (GPIO port output data register).
   ii. Explain how bit-banding improves GPIO control in real-time applications

[4] [CO01][BTL 3]

b. You are working with a 32-bit control register in the STM32F401CCU6 microcontroller. The register enables and disables multiple hardware features using different bits, and several bits are already set.
   i. Apply bitwise logic to write a C statement that disables only bit 5, without modifying the state of other bits in the register.
   ii. Compare this approach with the use of a direct assignment such as REG = 0;.

[4] [CO01][BTL 3]

PART B (Practical)[30 Marks]

6) Design and implement a **Smart Lighting with Door Sensor and Manual Override system** on an STM32F401CCU6 microcontroller. The hardware setup is as follows: switches are connected to PA0 (door sensor) and PB10 (manual override), an LDR is connected to channel 1 through the ADC, and LEDs are connected to PC13, PC14, and PB9. The system should work as follows:

a. when the switch on PA0 is pressed, it simulates a door opening, and LED PC13 must turn ON, while releasing the switch (door closed) should turn it OFF.
b. The LDR on PA1 is used for automatic lighting: when the ambient light is low (ADC value - set threshold), LED PB9 should turn ON, and when the light is high, it should turn OFF.
c. In addition, the switch on PB10 acts as a manual override for LED PB9; pressing it should temporarily disable the auto-light function and toggle the LED ON or OFF with each press, resuming automatic LDR control once override is released.

Rubrics:

Register Configurations	[10] [CO02][BTL 3]
Logic Design	[5] [CO02][BTL 4]
Implementation	[5] [CO02][BTL 4]
Interfacing the Hardware	[5] [CO02][BTL 4]
Output	[5] [CO02][BTL 4]

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**20**	BTL 1	
CO02	**30**	BTL 2	**12**
CO03		BTL 3	**18**
CO04		BTL 4	**20**
CO05		BTL 5	
CO06		BTL 6	

Bloom's Taxonomy Levels (attached for reference)

Level 1 – Remember

Level 2 - Understand

Level 3 – Apply

Level 4 - Analyze

Level 5 - Evaluate

Level 6- Create

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Answer all questions

PART A (Theory)[20 Marks]

1) You are helping your team choose the right processor for two different devices: A smartwatch that needs to run on battery for many days. A desktop computer that runs heavy software like video editors and games. Your mentor mentions two types of processor designs: RISC (used in ARM Cortex) and CISC (used in Intel processors).
   Based on this scenario: Explain the difference between RISC and CISC by comparing the ARM Cortex processor (used in the smartwatch) and a typical Intel processor (used in the desktop).
   [2] [CO01][BTL 2]
2) Explain the functionalities of ARM Cortex M4 Core registers. [4] [CO01][BTL 2]
3) How does the ARM Cortex-M4 use the Link Register (LR) during subroutine calls?
   [3] [CO01][BTL 2]
4) What information is stored in the Program Status Register (PSR) of the ARM Cortex-M4 processor? Explain the purpose of the condition flags used in decision making.
   [3] [CO01][BTL 2]

5)

a. Apply the concept of bit-banding in ARM Cortex-M4 to control a GPIO pin using STM32F401CCU6.
   i. Calculate the bit-band alias address for bit 6 of the register located at 0x40020014 (GPIO port output data register).
   ii. Explain how bit-banding improves GPIO control in real-time applications.

[4] [CO01][BTL 3]

b. You are configuring a GPIO pin using a 32-bit register in STM32F401CCU6. Initially, some other bits in the register are already set. Write code to enable bit 6 without affecting the other bits, and explain why your approach is safer than using direct assignment. Support your explanation by comparing it with a case where REG = (1 << 6); is used.

[4] [CO01][BTL 3]

PART B (Practical)[30 Marks]

6) Design and implement a **Smart Fan Speed and Status Indicator system** on an STM32F401CCU6 microcontroller. The hardware setup is as follows: a switch is connected to PA6 to act as the fan power control, a potentiometer is connected to ADC1 channel 0 to regulate fan speed, another switch is connected to PB10 for manual override, and LEDs are connected to PC13, PC14, and PB9 for system indication. The system must operate in the following manner:
   a. when the switch on PA6 is pressed, it simulates turning the fan ON and the LED on PC13 should glow, while releasing the switch turns the fan OFF and the LED goes OFF.
   b. The potentiometer reading determines the fan speed level, which should be indicated by the blinking frequency of PB9 with two bands only: slow blinking for low speed and fast blinking for high speed.
   c. The manual override switch on PB10, when pressed, should force PB9 to stay steadily ON regardless of the potentiometer input, and simultaneously PC14 should glow steadily to indicate that override mode is active. Releasing PB10 must resume potentiometer-based automatic fan speed control. Importantly, if the fan is OFF (PA6 not pressed), both PB9 and PC14 must remain OFF irrespective of potentiometer or override input.

Rubrics:

Register Configurations	[10] [CO02][BTL 3]
Logic Design	[5] [CO02][BTL 4]
Implementation	[5] [CO02][BTL 4]
Interfacing the Hardware	[5] [CO02][BTL 4]
Output	[5] [CO02][BTL 4]

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